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Field Oriented Control of Tubular PM Linear Motor Using Linear Hall Effect Sensors

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Abstract— **A method to determine the shaft position of a tubular linear permanent magnet motor is presented. Instead of using an incremental optical or magnetic encoder, linear Hall Effect sensors could be used to determine the exact shaft position. This will reduce the cost, size and remove the trailing sensor cable. A Phased Locked Loop was proposed to reduce the noise in the signals. Experimental results proved the effectiveness of the presented method in Field Oriented Control.**

Keywords-component; Hall effect devices, Motor drives, Permanent magnet motors, Phase locked loops.

I. NOMENCLATURE

v	phase voltage.
i	phase current.
R	phase resistance.
L	synchronous inductance.
λ	flux linkage.
ω	angular electrical speed.
u	linear speed.
θ	angular electrical position.
x	linear position.
τ_p	pole pitch.
F	electromechanical force.
$\alpha\,\beta$	stationary reference frame.
$d\,q$	rotating reference frame.
$\hat{\cdot}$	estimated value.

II. INTRODUCTION

Permanent Magnet (PM) Linear Motors are being used in various applications which ranges from Maglev trains to factory automation where it started to replace many mechanical linear motion systems including ball-screws and pneumatics [1, 2]. Linear motors could be driven using same control methods developed for rotating motors including: (1) Six-step operation, (2) Conventional Direct Thrust Control (DTC), (3) modified Direct Thrust Control [3, 4], and (4) Field Oriented Control (FOC). To avoid the high force ripple, FOC was implemented in this paper.

The cost of these motors remains a barrier against a broader implementation. Many attempts have been made to reduce the cost or extend the operation of linear motors. In [5], the stroke was increased by running the motor while the PM is partially outside the armature winding. A sensorless control method was proposed for conventional DTC in [6] to eliminate position sensors. DTC uses the speed feedback but does not require an accurate position feedback.

Field Oriented Control requires a more precise and accurate position feedback which is usually achieved by an incremental optical or magnetic encoder using a strip scale.

Although these linear encoder are proven to be effective and highly reliable, they have their own problems. The high cost, difficulty of installation, the trailing sensor cable, the need for offset calibration and the unknown position at start-up may create a big challenge to implement them in real life applications. An alternative solution is to use the less expensive linear Hall sensors. In this case, two Hall elements are required where the first is placed in phase with one winding and the second is shifted by 90 electrical degrees. If the motor windings are designed to be stationary with a moving magnet, then the sensor will be stationary too. Since the air gap flux is sinusoidally distributed, the output of two sensors will be a sine and cosine functions of the position. Using an Analog to Digital Converter (ADC), these two signals could be supplied to a microcontroller where they can be directly used for Park and inverse Park transformations. This solution may not be practical in real life applications due to the inevitable noise in the measurements. In this paper, a Phase Locked Loop (PLL) is used to extract the position information from these signals. Linear Hall sensors were implemented to drive a rotating PM motor directly in [7] and using PLL in [8].

Section II of this paper presents the model of PM linear motor and describes the FOC. In Section III, the Hall sensors along with a PLL method to calculate the position and speed is presented. In Section IV, experimental results showed the effectiveness of using Hall sensors with PLL compared to the incremental encoder under FOC.

III. FIELD ORIENTED CONTROL OF PM LINEAR MOTOR

In this paper, a tubular PM linear motor was employed although the same discussion could be applied to flat and U-shaped linear motors. The motor model in the synchronous reference frame is given by [1, 2],

$$\begin{bmatrix} v_d \\ v_q \end{bmatrix} = \begin{bmatrix} R + sL_d & -\omega L_q \\ \omega L_d & R + sL_q \end{bmatrix} \begin{bmatrix} i_d \\ i_q \end{bmatrix} + \begin{bmatrix} 0 \\ \omega \lambda_{pm} \end{bmatrix} \tag{1}$$

The relationship between the angular and linear frames is given by,

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$$\omega = \frac{\pi}{\tau_p} u \tag{2}$$

And,

$$\theta = \frac{\pi}{\tau_p} x \tag{3}$$

Where x and u are the linear shaft position and speed respectively and τ_p is the pole pitch. The force could be derived as,

$$F = \frac{3}{2}\frac{\pi}{\tau_p}(\lambda_{pm}\, i_q + (L_d - L_q)\, i_d\, i_q) \tag{4}$$

To control the shaft position of the linear motor using Field Oriented Control, three closed loop control loops are required. The most inner loop is the current loop where two PI controllers are used to force the motor current to track the reference currents. Because there is no variation in the inductance of this motor as the position changes ($L_q = L_d$), a control scheme similar to Surface Mount PM motor could be implemented. The current I_d^* was commanded to be zero while the current I_q^* which represents the force command was set to be the output of the speed loop. Another PI controller is designed to close the speed loop. Finally, a position loop is formed where the output of the controller is employed as the command speed. Fig. 1 shows the control system complete block diagram.

IV. PHASED LOCKED LOOP

In this section, the method of measuring the shaft position using two linear Hall sensors with PLL is discussed. First, the two Hall elements were placed in a position where the stator magnetic field is negligible. This results in reduced stroke. Figure 2 shows the sensor placement where the first linear hall sensor was placed in phase with winding 'A' and the second sensor is placed after 90 electrical degrees so that they can provide the perfect sine and cosine signals with the same gain and no offset or saturation. However, the sine and cosine signals were too noisy to be employed directly in the closed loop system. Moreover, the speed calculation requires the derivative of these noisy signals which will amplify the noise.

As an alternative, the error in position was calculated and a PLL was formed to compensate for this error. The sine and cosine signals are basically the flux in the two axes of the stationary reference frame given in per unit. Transforming these signals to the natural reference frame will result in $\lambda_d = \lambda_{pm}$ and $\lambda_q=0$ as expected. The estimation process starts from an estimated frame with a position $\hat{\theta}(t)$ as shown in Fig. 3.

The error in shaft position is defined as,

$$\theta_{err}(t) = \theta(t) - \hat{\theta}(t) \tag{5}$$

Transforming the sine and cosine signals to the estimated reference frame will result in,

$$\begin{bmatrix} \hat{\lambda}_d \\ \hat{\lambda}_q \end{bmatrix} = \begin{bmatrix} \cos(\hat{\theta}) & \sin(\hat{\theta}) \\ -\sin(\hat{\theta}) & \cos(\hat{\theta}) \end{bmatrix} \begin{bmatrix} \cos(\theta) \\ \sin(\theta) \end{bmatrix} \lambda_{pm} = \begin{bmatrix} \cos(\theta_{err}) \\ \sin(\theta_{err}) \end{bmatrix} \lambda_{pm} \tag{6}$$

Then, the error between the real and estimated position could be derived as,

$$\theta_{err}(t) = \tan^{-1}\left(\frac{\hat{\lambda}_q}{\hat{\lambda}_d}\right) \tag{7}$$

Finally, a PI controller is designed to compensate for this error and get the position and speed information. The same structure has been proposed for position sensorless drives [9]. The output of this controller is the speed which could be integrated to get the position as shown in Fig. 4.

Compared to an incremental encoder, this approach will result in a less expensive and smaller size absolute motor shaft position sensor with a stationary cable.

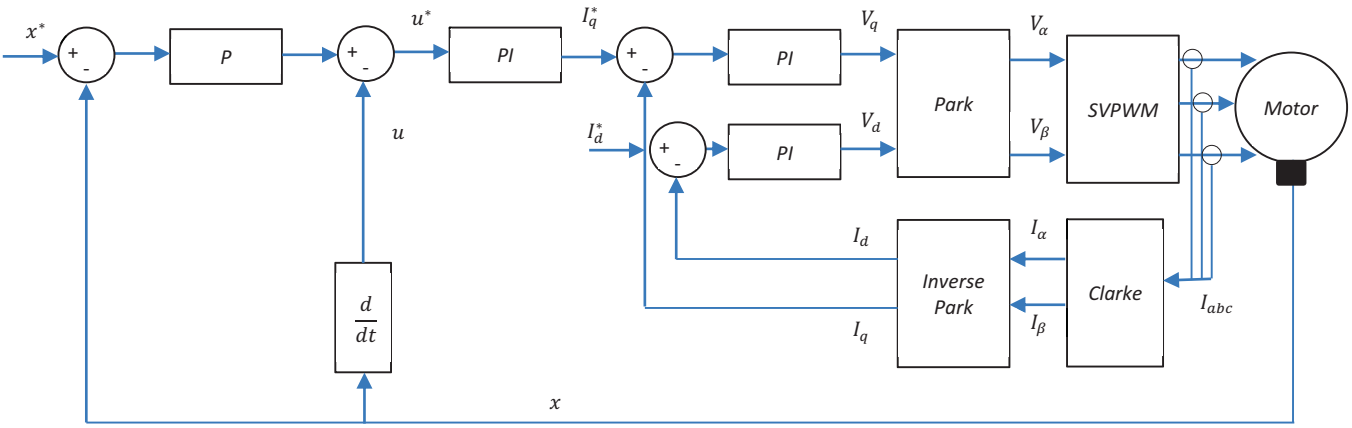

Fig. 1: Block diagram of Field Oriented control of the PM linear motor.

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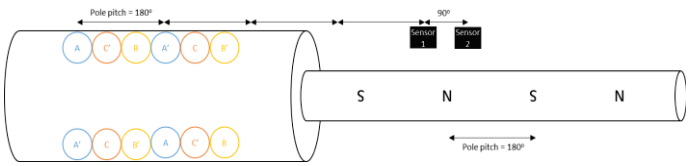

Fig. 2: Linear Hall sensor placement.

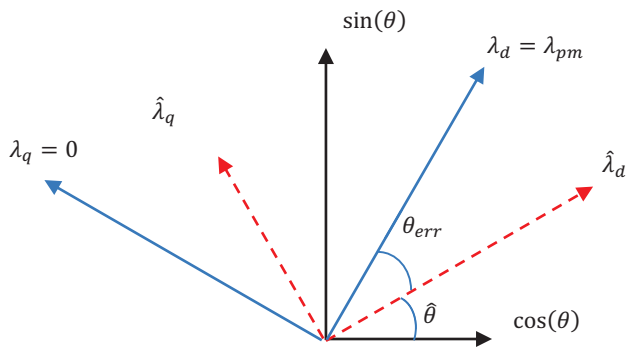

Fig. 3: Estimating the position error from sine and cosine signals.

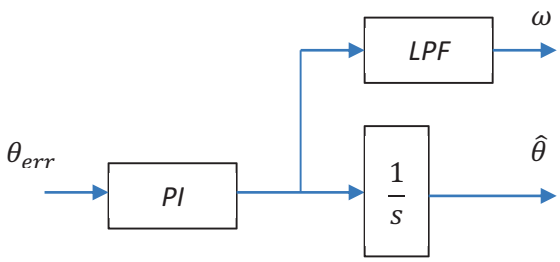

Fig. 4: Phase Locked Loop consists of a PI controller, an integrator and a low pass filter.

V. Sensor Misplacement Analysis

In this section, the effects of linear hall effect sensors misplacement is studied. In theory, three types of errors are possible: (1) If the two sensors are still 90° apart but both are shifted with respect to winding 'A' with a small offset, (2) The first sensor is placed perfectly in phase with winding 'A' but the second sensor is not exactly 90° apart, (3) One of the two sensors has a different gain.

In practice, any of these scenarios are possible. The worst case scenario will be a combination of these three. However, in this section each case is discussed individually.

To show the effect of the first case, the motor was assumed to run at 1 Hz and a 5° offset was introduced to both sensors. Since the error in position is constant, the force was reduced but no ripple was introduced as shown in Fig. 5.

Figure 6 shows the results of the second case where the motor is running at 1 Hz and a 5° offset was introduced only to the second sensor. In this case, a force ripple was resulted.

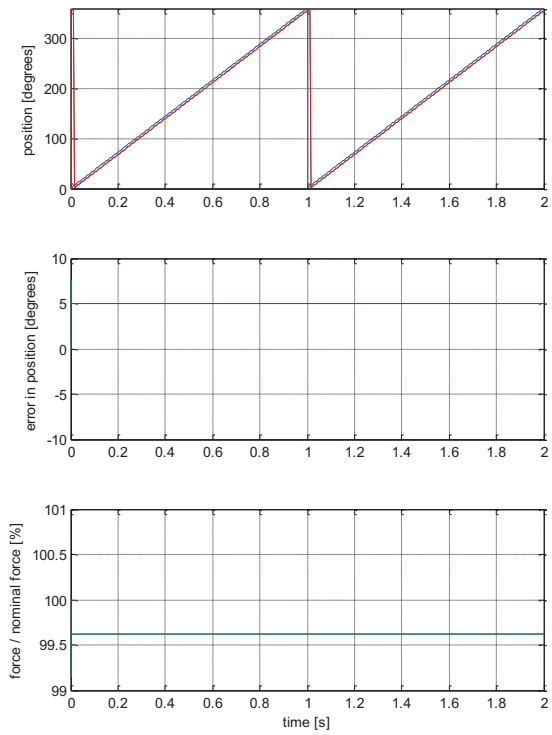

Fig. 5: (a) Real and measured motor electrical position, (b) Error in the position, (c) Percentage of resultant force compared to nominal force for the first case.

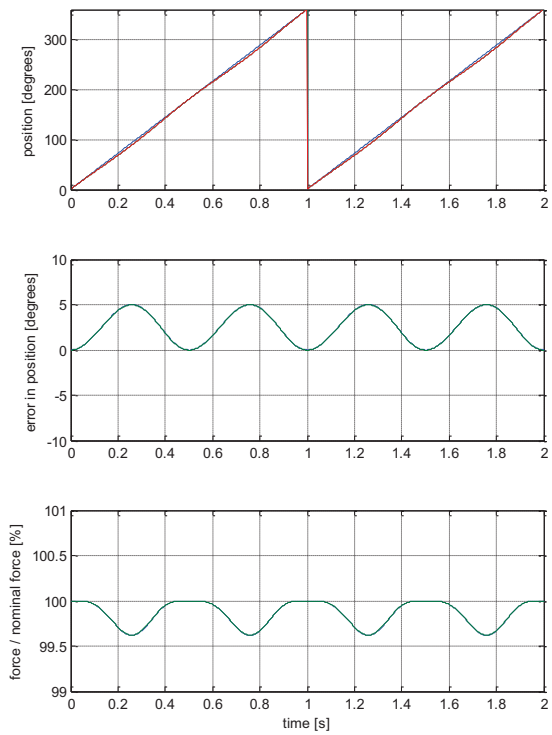

Fig. 6: (a) Real and measured motor electrical position (b) Error in the position (c) Percentage of resultant force compared to nominal force for the second case.

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For the third case, a 10% gain was added to the second sensor. Again a force ripple was observed as shown in Figure 7. In general, the ripple will be less because the error in position in this case happens in both directions where in the second case the error was only positive.

In the experimental test presented in the next section, misplacement error was still present despite the fact that great care was taken to place the sensors accurately. A combination of both the second and third case was detected. The two sensors where 89° degrees apart instead of 90° and the 3.5% difference in the gain of the two sensors was present. The Analog to Digital Converter (ADC) gain was adjusted to compensate for the sensors gain difference. Based on the results of this section, the effects of 1° offset misplacement was assumed to be negligible.

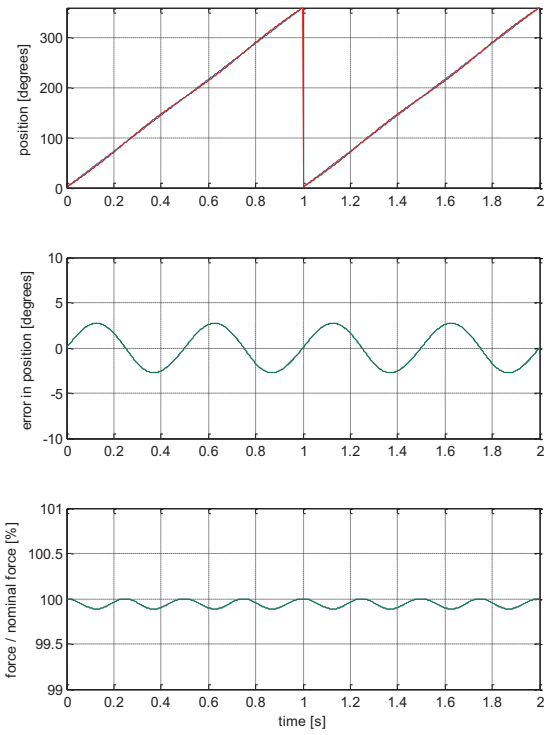


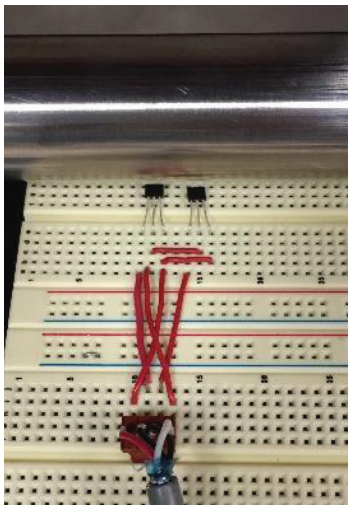
Fig. 7: (a) Real and measured motor electrical position, (b) Error in the position, (c) Percentage of resultant force compared to nominal force for the third case.

VI. EXPERIMENTAL RESULTS

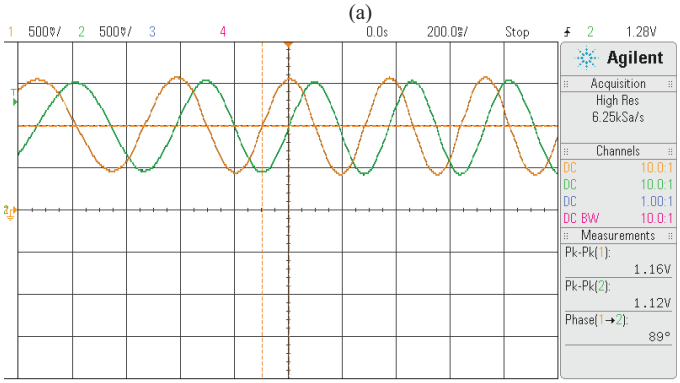
Position FOC was implemented using two different position sensors. An incremental magnetic encoder with a resolution of 5 µm was used to detect the correct position for the purpose of comparison and evaluation only and was not used to control the motor. Two DRV5053 analog bipolar Hall Effect sensors connected to a 12-bit ADC were used to provide the motor shaft position feedback to the control loop.

A dc bus voltage of 60 V along with a three phase IGBT inverter switching at 10 kHz were used to drive the three-phase Tubular PM linear motor. The winding assembly was mounted to the base while the magnetic rod was allowed to slide freely and drive the load.

The motor was loaded using a Trans-Rotary Magnetic Gear [10]. The load force was selected to be half of the rated continuous motor force. In Fig. 8, the Hall sensors and their output signals are shown while the experimental setup used in this paper is shown in Fig. 9.



(a)



(b)

Fig. 8: (a) Prototype showing two DRV5053 Hall sensors placed 90 degrees apart under the magnetic shaft, (b) The PM flux distribution measured by the two Hall sensors as the shaft moves.

The position and speed derived from the linear encoder were used to monitor the real values but were not utilized in the control. Fig. 10 compares the position derived from the two sensors while the position and speed derived from Hall sensor is used to close the three loops.

Fig. 11 shows the speed derived from both methods. From Figs. 10 and 11, it is clear that there is a good agreement between the results obtained from the two methods. In Fig. 10, there is an offset in the position derived from the Hall sensors. The maximum error was around 4 electrical degrees. In Fig. 11, the speed derived from the Hall sensor had more ripple.

The motor was commanded to move a stroke of 400 mm. When the incremental encoder was used to close the loops, the motor took 5.8 seconds to arrive at the commanded position. Due to the small error in the estimation, the motor took 6.3 seconds when the Hall sensors were employed.

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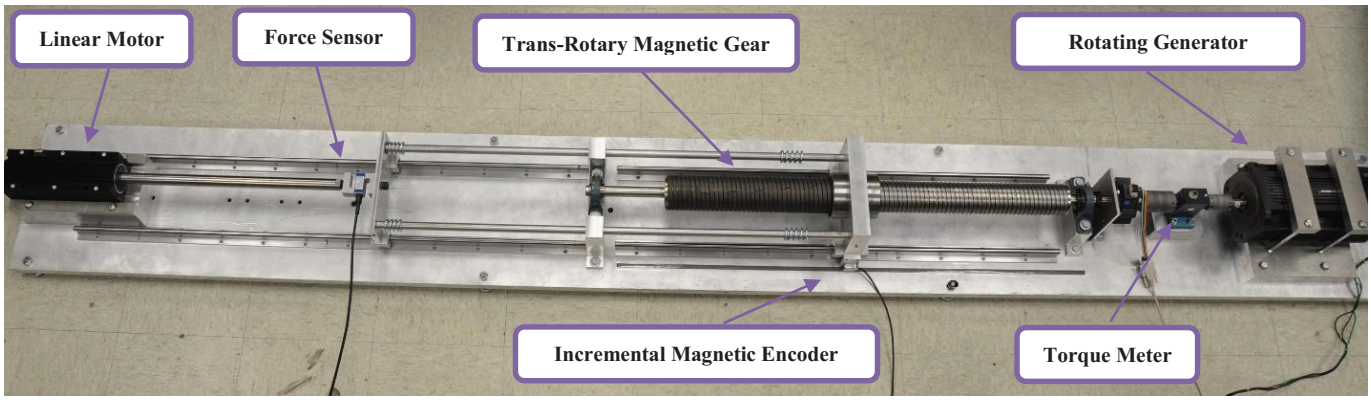

Fig. 9: Experimental setup including the linear motor, Trans-Rotary Magnetic Gear and rotating generator.

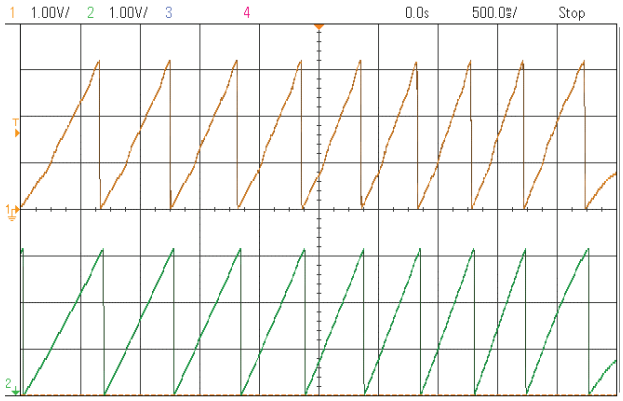

Fig. 10: Position [2 rad/div] measured using: Hall sensors and PLL (top), incremental encoder (bottom).

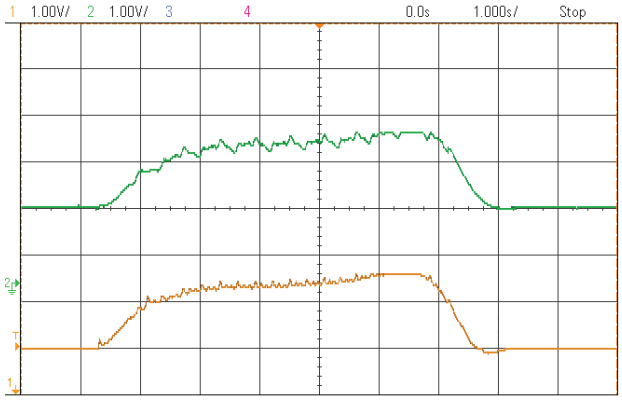

Fig. 11: Speed [50 (mm/s)/div] measured using: Hall sensors and PLL (top), incremental encoder (bottom).

VII. CONCLUSIONS

This paper shows that two Hall Effect sensors can provide sine and cosine signals which may effectively be used to provide an accurate position information to drive a tubular PM linear motor using Field Oriented Control. This will eliminate the high cost and difficulties associated with the use of linear optical or magnetic encoder. A Phased Locked Loop system which utilizes the sine and cosine signals to estimate the speed and position was presented. The effect of misplacement of the Hall Effect sensors was discussed. Finally, the proposed method was experimentally verified.

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